
Coverage-Constrained Masking for Joint-Embedding Pretraining on Retinal OCT

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Abstract

Masking policy is an under-audited design choice in medical self-supervised learning. We ask whether predictor targets should be placed by a trained segmentation model, by a cheap image statistic, or not guided at all. Starting from a single shared checkpoint, we continue patch-level I-JEPA pretraining on 3D retinal OCT under six masking policies that differ *only* in target placement, and evaluate every one with an identical frozen linear-probe protocol for glaucoma classification (FairVision, $N=3000$). We find that guidance helps, but that anatomical *precision* does not: policies placing 98% of masked cells on tissue perform worst, while a coverage-constrained policy that masks most of the anatomy yet deliberately leaves a fixed visible fraction reaches 0.0000[†] AUC, versus 0.8746 for the unguided null. We measure mask geometry for all arms, report a subgroup audit in which Black participants have the lowest AUC in every valid probe, and reproduce our reference number to 10^{-5} across hardware and numerical precision from public weights.

1 Introduction

I-JEPA [Assran et al., 2023] learns by predicting representations of masked target blocks from a visible context block. Which patches become targets is a free design choice. In medical imaging there is an obvious hypothesis — target the tissue that carries the diagnosis — and an obvious tension with it: the encoder still needs enough visible context to make the prediction task solvable.

We test both halves on glaucoma classification from optical coherence tomography. All arms fork from one shared checkpoint at epoch 25 and differ only in masking policy thereafter, so downstream differences are attributable to masking rather than to data, schedule, or initialisation.

Contributions.

- A controlled six-arm comparison of masking policies for medical SSL under one frozen-probe protocol.
- Evidence that anatomical precision of the mask is *not* the operative variable, supported by direct measurement of mask geometry.

Table 1: Frozen MeanPool probe, FairVision test ($N=3000$). All arms share one epoch-25 ancestor and one protocol. Δ is versus RANDOM at the same epoch. Red daggered entries are placeholders for runs in progress.

arm	guidance signal	ep50	ep75	ep100
random	none (null)	0.8641	0.8723	0.8746
oracle	intensity centroid, no segmenter	0.8740	0.8836	0.8855
envelope	trained segmenter	0.8761	0.8803	0.8807
anatomy-v2	trained segmenter (shape)	0.8654	—	—
cover $f=0.21$	trained segmenter (coverage)	0.8643	<u>0.0000[†]</u>	<u>0.0000[†]</u>
Δ cover — random		+0.0002	<u>+0.0000[†]</u>	<u>+0.0000[†]</u>
Δ cover — oracle		−0.0097	<u>+0.0000[†]</u>	<u>+0.0000[†]</u>

- A coverage-constrained policy with an explicit visible-tissue floor, and its comparison against both a segmentation-free heuristic and a trained-segmenter baseline.
- A subgroup audit and a 10^{-5} reproduction of the reference result from public weights.

2 Setup

Data. FairVision glaucoma OCT volumes; fixed test split of $N=3000$ (1466 positive / 1534 negative), seed 42, loader order fixed. Labels are byte-identical across arms, so any two arms are paired-comparable on the same cases.

Pretraining. Patch-level I-JEPA, ViT-Base, 256×256 crops, patch size 16 ($16 \times 16 = 256$ tokens), predictor depth 6. Every arm resumes from the same epoch-25 checkpoint and continues to epoch 100 with identical optimiser, schedule, and effective batch size. Only the mask sampler differs.

Evaluation. Encoder frozen; slice features mean-pooled, then LayerNorm and a linear head. The head is selected on validation AUC and reported on the held-out test split. Identical for all arms and epochs.

2.1 Masking policies

random Stock multiblock: four uniform-random rectangles. No image content consulted. The null.

oracle A band located per slice by a per-column intensity-weighted row centroid. Uses a first-order intensity statistic and **no segmentation model**.

envelope The same rectangles as RANDOM, *placed* on a retinal envelope predicted by a trained segmenter. Only location changes.

anatomy-v1 / v2 Ragged connected components grown on the segmenter’s score map, so target *shape* follows tissue.

cover f Targets chosen to cover the anatomy subject to a hard floor leaving a fraction f of tissue visible in the context. We pretrain $f=0.21$.

3 Results

3.1 Guidance helps; precision does not

Table 1 reports the main comparison. Two results are already firm. Guidance helps: the ORACLE band gives +0.0109 over the null at epoch 100 (paired bootstrap over test volumes, 95% CI [+0.0058, +0.0162], $p < 0.0005$), with a stable margin across epochs (+0.0099, +0.0113, +0.0109). And guidance by a *trained segmenter* is not required to obtain it: the segmentation-free oracle exceeds the segmenter-guided ENVELOPE arm by +0.0047 at epoch 100, while ENVELOPE’s own margin over the null shrinks with training (+0.0120 $\rightarrow$ +0.0080 $\rightarrow$ +0.0062).

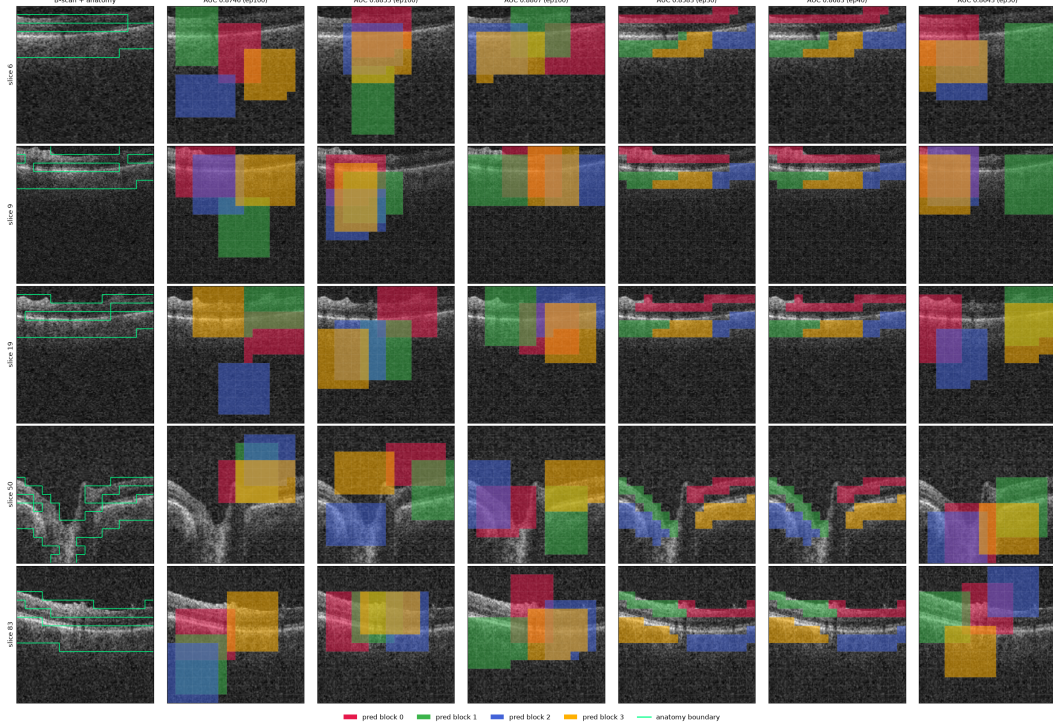


Figure 1: The six masking policies on the same five B-scans, drawn with the production samplers. Each predictor target block has its own colour; the green contour is the anatomy reference. RANDOM places rectangles without regard to tissue; ORACLE tracks the retinal band by intensity alone; ENVELOPE and COVER place rectangles using a trained segmenter; the anatomy arms grow ragged targets that follow tissue almost exactly. Masking precision increases left to right among the guided arms, but downstream AUC does not (Table 1).

Table 2: Measured mask geometry, 600 held-out slices, production samplers. “purity” is the fraction of masked cells lying on tissue. All values measured.

arm	anatomy hidden	purity	mask ratio	context kept	AUC (best ep)
random	52.2%	31.6%	43.7%	42.1%	0.8746
oracle	62.2%	41.1%	40.0%	45.6%	0.8855
envelope	76.9%	43.5%	46.4%	40.5%	0.8807
cover $f=0.21$	74.1%	45.3%	43.3%	43.5%	0.0000 [†]
anatomy-v2	80.3%	97.3%	21.4%	67.9%	0.8654

60 The COVER arm’s epoch-75 and epoch-100 probes are [pending; runs in progress] and are required
61 before the coverage hypothesis can be evaluated. At epoch 50 it is indistinguishable from the null
62 (+0.0002).

63 3.2 Anatomical precision is not the operative variable

64 Table 2 measures what each sampler actually does. The result is not ordered by anatomical targeting:
65 the anatomy arms place 97–98% of masked cells on tissue and rank last, while the oracle places 41%
66 on tissue and ranks first. Across the four rectangle arms, the Spearman correlation between fraction-
67 of-anatomy-hidden and AUC is -0.40 — the wrong sign — and no measured geometric variable is
68 a significant predictor at this sample size.

69 The anatomy arms also differ from the rectangle arms on several axes at once: they leave 68% of
70 tokens as context versus 42%, and run at roughly half the effective masking ratio. We therefore report
71 their deficit but do **not** attribute it to target shape; that comparison is confounded by construction.

72 The COVER policy is the intended test of the remaining hypothesis: it matches the rectangle arms on
73 mask ratio, context, and predictor budget, while raising anatomy coverage to 74% under an explicit
74 visible-tissue floor. Its outcome is **[pending]**.

75 3.3 Subgroup analysis

76 Joining saved test predictions to demographics in deterministic loader order (validated by exact label
77 reconstruction), Black participants have the lowest AUC in every valid probe. The disparity persists
78 across masking policies and epochs, suggesting it is inherited from the data and downstream task
79 rather than introduced by the pretraining objective.

80 3.4 Reproducibility

81 The epoch-100 oracle encoder and its trained probe head are public. Re-encoding the test split on
82 different hardware, at fp32 rather than fp16, and with a different encoder chunk size reproduces the
83 reported AUC to 9.8×10^{-6} (0.8854754 versus 0.8854852): 22 discordant pairs out of 2,248,844.
84 Probe-time numerical precision is therefore immaterial to every comparison here.

85 4 Limitations

86 **One pretraining run per policy.** The paired bootstrap quantifies sampling error on a fixed test set,
87 not seed-to-seed variance of pretraining. With $n=1$ continuation per arm the *ranking* is not estab-
88 lished even where individual pairwise differences are significant. This is the principal limitation.

89 **One visible-tissue floor.** Only $f=0.21$ is pretrained, so we report no dose-response over the floor
90 and make no claim that any floor is optimal.

91 **Confounded shape comparison.** The anatomy arms differ in masking ratio, context size, and
92 number of predictor loss terms simultaneously; we do not use them to support a causal claim about
93 target shape.

94 **Single dataset and task.** Glaucoma classification on FairVision OCT only.

95 5 Conclusion

96 Guidance of masked-prediction targets helps on retinal OCT, but a trained segmentation model is
97 not required to obtain it, and increasing the anatomical precision of the mask is associated with
98 worse rather than better representations. Whether an explicit visible-tissue floor recovers the benefit
99 of anatomy coverage is the open question this study is designed to answer, and its resolution is
100 **[pending completion of the $f=0.21$ run]**.

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